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Bike-sharing or taxi? Modeling the choices of travel mode in Chicago using machine learning

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ABSTRACT

In many big cities, the bike-sharing system (BSS) and taxi play critical roles in transportation services. They both offer on-demand transportation options and allow flexible riding scheduling and routing. Previous literature has compared BSS and taxi to other transport modes, such as public transit and private automobile, but little is known about the spatiotemporal factors that influence travel choices between these two alternatives. Understanding travel patterns of BSS and taxi is critical in traffic demand analysis and sustainable transportation planning. Also, an in-depth examination of the patterns of travel behaviors, especially when one would choose BSS over a taxi, will provide valuable insights on human mobility and active living research. In this study, we investigated the spatiotemporal patterns of BSS and taxi trips in Chicago from 2014 to 2016. To model travel choices between BSS and taxi, we applied machine learning techniques to simulate the means of transport based on environmental and temporal factors. Results show seasonal trip variations of the BSS and a declining trend of taxi trips. BSS speed is relatively stable while taxi speed varies primarily because of time and locations. Based on the random forest model, which has demonstrated the best fit with high processing speed, travel distance and the number of parks and recreational facilities seem to be critical spatial predicting factors of the travel choice. Given any time and location, the model can recommend the travel choices between BSS and taxis for users. This study shows the significance of machine learning techniques in urban mobility research. Results of the study may potentially support people's transportation decision-making and facilitate sustainable transportation planning.

1. Introduction

Over the past two decades, cities around the world have engaged in developing sustainable transport systems. An emerging discussion of transport sustainability revolves around monitoring and modeling the mobility choices and predicting diverse mobility needs. Although studies have reported internal and external factors influencing transit ridership, they have primarily discussed public transport patronage in the context of a transit-private vehicle dichotomy, acknowledging that private vehicles provide spatial and temporal flexibility which fixed-route transit services cannot match. Overall, alternative transport modes and emerging systems that contribute to the integrated transportation network have not received adequate research attention.

One such alternative transport is a taxi. A taxi offers a speedy and convenient service to passengers. Although it has been criticized for competing for resources with public transit and giving rise to congestion and sedentary lifestyles, it complements public transport systems by fulfilling unique transportation needs. For example, taxi services provide spatial and temporal flexibility, especially for areas where access to public transit is low. As a result, it allows demographic groups without private vehicles to access locations without public transport. Furthermore, it is a crucial mobility option for specific population groups such as people with disabilities. Research has also suggested that people often combined taxi rides with public transportation (King et al., 2012). Despite the impact of emerging ride services (e.g., Uber and Lyft), taxi still plays an essential role in urban transportation systems. Literature has pointed out that there has been limited analysis of taxi compared to other transport modes in academic literature (Cooper and Mundy, 2016).

The bike-sharing system (BSS), on the other hand, enhances individual mobility, reduces congestion, and provides several additional benefits such as physical activity and emission reduction (Shaheen

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et al., 2013; Shaheen et al., 2014). Similar to taxis, people can use such means of transportation without the cost for maintenance and responsibility of ownership. The flexibility of BSS provides alternative means of transportation, especially for short-term trips (Shaheen et al., 2010). In the United States from 2001 to 2009, about half of all trips were shorter than 3 miles, according to the National Household Travel Survey (Flusche, 2016). However, more than 70% of these trips were driven (Darren, 2015). The BSS offers opportunities to reduce automobile-dependent trips and contributes to environmental and public health benefits (Faghih-Imani et al., 2014). Recent technological advancement has allowed "smart" bike-sharing where real-time bike availability information can be accessed, further contributing to the efficiency and convenience of the system. Because of its benefits, in recent years, BSS has been implemented in cities across the world, e.g., North America, Europe, and Asia, to support multimodal transportation.

Despite the differences between BSS and taxi, both transport modes offer unique flexibility concerning routing and scheduling and therefore are often considered viable choices to urban traveling. When it comes to the efficiency of these two means of transportation, it is traditionally believed that taking a taxi is much faster than riding a bike. However, the travel time and trajectory of taxis vary significantly according to the spatial characteristics of a city and the availability of other transport options. Recently, studies have compared the efficiency of these two means of transportation. Jensen et al. (2010) analyzed the Lyon BSS data and showed that riding a bike can compete with cars regarding speed in core downtown areas, particularly during peak hours. Faghih-Imani et al. (2017a) found that in New York City, BSS is either faster or competitive with taxi during weekdays' morning, mid-day, and afternoon periods.

With the benefits of emission reduction, reduced congestion, health, and well-being, if bike sharing trips can substitute taxi trips, there will be significant policy implications. Research has discussed the roles of bike sharing in substituting private car journeys in a variety of contexts (Fishman et al., 2014a; Murphy, 2010). However, little is known concerning the factors that explain an individual's travel mode choices between taxi and bike-sharing. Without this knowledge, planners and policy-makers may miss critical opportunities to estimate the demand-supply dynamics related to taxi dispatch, promote BSS usage, and regulate and diversify urban transport means.

To fill this gap, the main objective of this paper is to model the travel mode choice between BSS and taxi, and how different spatio-temporal factors influence such a choice. Furthermore, we utilize multiple machine learning methods to predict spatiotemporal variations of travel patterns by external determinants, such as the time of a day, the direction of travel, and density of BSS stations and bike lanes. To the best of our knowledge, using BSS and taxi data concurrently and machine learning techniques to predict travel mode choices between the two modes while considering the influences of contextual attributes is very new in the literature. The findings would reveal key determinants of travel modes and potentially facilitate sustainable transportation planning. In the next section (Section 2), we will introduce related studies about BSS and taxi modeling and recognized factors influencing travel mode choices. Next, we will present the data and analytical methods in Section 3. In Section 4, we will discuss the spatiotemporal patterns of the two types of transport mode and the results concerning the contextual factors based on different machine learning techniques. Finally, we will summarize the main findings and discuss directions for future research in Section 5.

2. Related work

2.1. BSS and taxi data and their applications in understanding cities

Station-based BSS has been one of the most popular forms, with 36.5 million trips taken by people in the US in 2018 (NACTO (National Association of City Transportation Officials), 2019). Among the station-based BSS, the largest four systems – Citi Bike NYC, Divvy, Capital Bikeshare, and Hubway – generated the majority (74%) of all rides in 2017 (NACTO (National Association of City Transportation Officials), 2018). All these four systems and many others provide trip level data. On the other hand, taxi data is less available than BSS data, but a few cities provide such information, such as New York City and Chicago. Both cities provide trip records, including fields such as pick-up and drop-off time and locations, trip distances, and fares. BSS and taxi data provide important data source to better understand transportation patterns in cities.

Given the increasing availability of citywide trip and prosperity of human mobility research, researchers have used BSS and taxi data to investigate many urban issues, such as city structure and transportation congestion. City structure is primarily associated with travel patterns of citizens. Therefore, by analyzing traffic flows and mobility patterns, we can discover the "skeletons and pulses" of cities, such as land use condition, resource distribution, and social events (Liu et al., 2012). Liu et al. (2015) summarized a two-level hierarchical polycentric city structure in Shanghai based on modeling intra-city taxi flow data. The street hierarchies were classified based on network analysis and large-scale taxi traffic flow in Wuhan, China (Huang et al., 2016). Chen et al. (2015) applied anomaly detection technique and heuristic modeling to detect events and urban activities based on the BSS data. This method allowed them to identify 89% of the social activities in the major urban centers of Washington, D.C.

Another emerging strand of research is related to tackling urban congestion problems with BSS and taxi data, where data-driven approaches are often used to analyze and predict traffic conditions. Meng et al. (2017) modeled the traffic volume on road segments based on data collected by loop detectors and taxi trajectories in Guiyang, China. In addition to daily congestion patterns, some non-recurrent congestion may be caused by accidents or newly established building sites. By analyzing taxi data, these non-recurrent congestions can be revealed. Keler et al. (2017) extracted and compared the recurrence and non-recurrence traffic congestion based on massive Floating Taxi Data (FTD) set in Shanghai (Keler et al., 2017). As for BSS, BSS tend to be regarded as a cure for traffic congestion. Several studies have investigated the effects of BSS on congestion reduction. Hamilton and Wichman (2017) examined the impact of bicycle-sharing infrastructure on urban transportation. Results suggest that the availability of BSS can reduce traffic congestion. Such an effect is especially concentrated in highly congested areas. Wang and Zhou (2017) examined the influence of BSS on congestion in 96 US urban areas across ten years, where a mixed effect of BSS on congestion was found. Although there have been several studies investigating city structures and congestions based on BSS and taxi data, few prior studies have explored the spatiotemporal factors associated with BSS and taxi usage and preference.

2.2. Factors associated with BSS and taxi usage and preference

There are a number of external factors associated with BSS and taxi usage. Regarding the demand for a taxi, previous studies have revealed socio-demographic and environmental conditions as essential factors in predicting trip numbers. Historically, taxi demand was estimated based on the total population of administrative districts. However, population-based demand analysis or permit processes has proven to be a poor indicator of the actual demand of a community (Cooper and Mundy, 2016). Schaller (2005) proposed a seven-factor model to predict taxi demand, including city size, availability of private vehicles, use of public transport, costs, service quality, competing modes and special populations. Recently, more studies have used actual taxi trip data to determine factors influencing the demand. For example, Qian and Ukkusuri (2015) investigated the spatial variation of urban taxi ridership of New York City based on geographically weighted regression and revealed that the urban form and access to subways had significant impacts on taxi ridership. Another study examining taxi trips in Washington D.C. confirmed that taxi trip demands were connected to land use patterns and accessibility to other modes of travel (Yang et al., 2018). Although these studies provided a foundation for examining factors influencing taxi trips, they discussed taxi trips alone and constructed models by aggregating data into spatial and temporal segments.

Compared to taxi models, BSS is an emerging topic in urban transportation literature. Inspired by Paris's BSS program (Nair et al., 2013), BSSs have been adopted by a wide range of cities over the world, and research on the factors associated with usage has been on the rise. For example, A Canada-based study showed that proximity to docking stations and membership status, as well as maintenance to be critical facilitators of BSS usage (Bachand-Marleau et al., 2012). Notably, living within 500 m of a docking station was related to a three-fold increase in one's odds of being a BSS user (Bachand-Marleau et al., 2012). Similarly, longer distances to docking stations from home, work, and other frequent destinations made it less convenient to riders and thus was the main barrier to BSS use (Fishman et al., 2014b). Land use and built environment variables, as well as their interactions with the time factor, were also demonstrated to influence bicycle flows (Faghih-Imani et al., 2014). In addition to land use and bike infrastructure, weather was identified to be crucial in influencing the demand for bike sharing (El-Assi et al., 2017).

Limited knowledge has been generated concerning using BSS as a substitute for a taxi in travel decision making. Nevertheless, there are some commonalities of BSS and taxi as transport modes in the urban transportation system. For example, BSS and taxi both offer on-demand transportation, and show short-trip competitive advantages compared to other on-demand modes such as limousines. They also allow flexible riding scheduling and routing (Faghih-Imani et al., 2017b). Considering a specific origin-destination pair, one may choose between taxi and BSS based on the proximity of taxi stands and bike dock stations, costs, speed, and delays, as well as other perceived contextual factors. Faghih-Imani et al. (2017) compared the travel time between BSS and taxi and demonstrated that bike share is faster or equally efficient for origin-destination pairs during specific temporal dimensions. Compared to taking a taxi, biking provides an alternative that shows environmental, social, and health benefits. Therefore, understanding the conditions under which a person would likely choose to use bike sharing rather than ride a taxi has critical implications for transportation planning and policy.

However, to date, very few pieces of research have examined the travel mode choices and preferences between BSS and taxi in cities, or factors that can predict such choices. To fill these gaps, we aim to construct the space-time typology of trips by BSS and taxi within different periods of a day and use different machine learning models to predict travel mode choices. Our findings aim to provide insights on the dynamics of the relative demand between BSS and taxi. The results regarding factor importance in predicting traveler's choices can potentially serve as a foundation for sustainable transportation planning and promotion. Also, patterns of travel behaviors with BSS and taxi will provide valuable references for human mobility research.

3. Data and methods

3.1. Data cleaning and preparation

We investigated the spatiotemporal patterns of bike sharing and taxi trips in Chicago from 2014 to 2016. Chicago's BSS, namely Divvy Bikes (https://www.divvybikes.com/), was launched in June 2013. When the system was initiated, it had about 75 stations and 750 bikes. The system expanded rapidly and it has more than 580 stations and 5800 bikes currently. Each station has a touchscreen kiosk which supports bike check-in and check-out using a member key or ride code. Data used in this study was obtained from the Divvy website. The dataset included both trip and station information in the past years. Trip data contained information such as origin and destination stations, start and end time of trips, trip duration, and user types. Station data consists of coordinates of the stations as well as dock capacities.

We performed data cleaning to remove records that were incomplete or outside of the scope of this study. Specifically, BSS trips that originated or ended at stations located outside the border of the city of Chicago were removed. As the BSS dataset only provided the trip duration, not the actual distance, we used the Google Maps Directions API to estimate the actual biking route between the origin and destination of each trip. Reasonable biking routes with road network distances were generated.

For taxi trips, in late 2016, the City of Chicago released the taxi data dating back to 2013. Over 100 million taxi trips were provided in the dataset. For each trip, information such as time trips started and ended, lengths of trips in time and distance, amount of taxi fare, starting and ending locations were reported. For privacy reason, the exact pickup and drop-off locations were masked off, only showing the Community Area and Census Tract. Also, all start and end times were rounded to the nearest 15 min. Taxi data was extracted from the Chicago Data Portal. Furthermore, we collected data regarding the built environment at the census tract scale to facilitate our modeling. For example, we obtained the total length of bike lanes (in km) downloaded from the Chicago Data Portal; and the number of parks and recreational facilities provided by the Chicago Park District (https://www.chicagoparkdistrict.com/).

Data were stored in a PostgreSQL database for data query and retrieval. We observed some abnormal records in the dataset and processed the data to remove the abnormalities. For instance, many trips had zero values for trip duration, but non-zero fares. In another instance, some trips originated and ended in different census tracts, but the trip distance values were zero. After consulting the team at the Chicago Data Portal, these records were very likely to be misreported in the system. We thus removed those records that show inconsistencies in the dataset. Similarly, when computing trip speeds, some trips had an average speed higher than 80 mph, which were unlikely to happen. Hence, trips with distance or duration as outliers were removed as well.

Comparing travel behavior based on BSS and taxi data at individual trajectory level would be ideal. However, due to the granularity of data from the taxi, we aggregated data to the census tract level. Specifically, taxi, BSS, and census data come in different spatial units; we took four steps to integrate them into a comparable framework. First, to explore the spatial patterns at the finest geographic scale possible, we removed the taxi trips that contained no census tract information. Second, to match the geographic scale, each BSS trip anchored by origin and destination was aggregated to the corresponding census tract. Third, to match the geographic extent, we also excluded trips that started or ended at the census tracts with no BSS stations. Fourth, the built environment data were also aggregated to each census tract, such as the total length of bike lanes and the number of park and recreational facilities.

3.2. Analytical methods

To describe the spatiotemporal patterns of the dataset, we analyzed the overall patterns of trip numbers, average speed, trip directions for BSS, and taxi trips. To account for the daily, seasonal, weekend vs. weekday variations, we aggregate the data into the respective time windows. We computed the mean travel speed for all trips between origins and destinations for BSS and taxi trips. Both trip numbers and travel speeds were compared from 2014 to 2016. In addition, to capture the daily temporal variations, we broke down the trips into seven different time periods based on trip start time, i.e., Night (0:00–7:00), Morning Rush (7:00–9:00), Morning (9:00–11:00), Lunch Break (11:00–13:00), Afternoon (13:00–17:00) Afternoon Rush (17:00–19:00), and Evening (19:00–24:00).

Trip directions reflected the dominant traffic pressures and city pulses. It varies across different time windows and locations. In this study, we computed trip directions based on centroids of census tracts. The computed angles were aggregated to eight-directional zones (each covering 45 degrees). We calculated the dominant trip directions for both BSS and taxi trips. Considering the bi-directional features of trips, trips with reversed start and end points (e.g., Trip 1 from A to B, and Trip 2 from B to A) canceled out. The resultant trip numbers from different census tracts were used to analyze the dominant trip directions.

To model the preference of BSS or taxi, we first prepared a number of extracted features, such as travel distance, time of day, travel directions, weather, and land use. Let $\{f_1, f_2, f_3 \ldots, f_n\}$ be a predefined set of classifiers, the goal is to train each classifier and find the best classifier that can predict travel mode preference based on given conditions.

We selected popular linear, nonlinear (not ensembles), and ensemble algorithms to model the preference. Different algorithms have their unique advantages. For linearly separable problems, linear models are fast and simple. Algorithms may have varied performance based on different nonlinear characteristics of feature space. Ensemble models combine several base algorithms to achieve optimal predictive model. We tested a few different algorithms under each category. Linear algorithms tested in this study include logistic regression, regularized linear models with stochastic gradient descent (SGD) learning. Nonlinear algorithms (not ensembles) include k-nearest neighbors (k-NN), support vector machine (SVM), gaussian naive bayes, decision tree (DT), and neural network.

k-NN assumes similar labels (or values) appear in close proximity. For classification tasks, an observation is classified by a majority vote of its neighbors based on certain types of distance function (e.g., Euclidean Distance). The choice of k depends upon the data distribution. Larger k values are usually associated with a smoother classification boundary. SVM aims to find a separating hyperplane that maximizes the distance between data points of different classes — finding this hyperplane help to achieve more confidence when classifying future data points. SVM can use kernels, certain mathematical functions (e.g., Radial Basis Function) to transform input data to a higher dimensional feature space so that a linear decision boundary can separate classes. SVM also use regularization parameter to control the tradeoff between the bias and variance of the model. Another supervised classification model used is naive bayes (NB). This model is a probabilistic method based on Bayes' theorem with the assumption of independence between features given the value of the class. Gaussian naive bayes is an extension of NB with real-valued features assuming a Gaussian distribution. DT splits training data into relatively homogeneous sets based on the most significant splitter for input features. To predict an observation, the model runs a sequential and hierarchical decision based on features to achieve predicted labels. A neural network is composed of an input layer, one or more hidden layers, and an output layer. Input and output layers are self-explanatory, usually holding features and predicted values, respectively. The hidden layer includes a certain number of neurons which are attached to activation functions. Deep neural network, in general, means a network with more than one hidden layer (Alpaydin, 2014; Caruana and Niculescu-Mizil, 2006). More in-depth description of each model is reviewed in previous literature (Kotsiantis et al., 2006; Sun et al., 2009).

Ensemble models combine the decisions from multiple models to improve the final performance. It usually includes two general approaches, bagging and boosting. Ensemble models used in this study include AdaBoost, Bagged Decision Trees, Random Forest, Extra Trees, and Gradient Boosting Machines. Among these models, Random Forest, Extra Trees, and Bagged Decision Trees are considered as bagging approaches. Bagging usually samples training data set with replacement. Then, a model is created for each sample. Results of these multiple models are combined using the most voting for the final predictions. For instance, the random forest builds many trees, and it makes a prediction based on the most voting results. AdaBoost and Gradient Boosting are considered as boosting approach. Boosting is a sequential approach in which, the first model is trained on the entire data set, and the subsequent models are built by fitting the residuals of the previous models. Such iterations allocate higher weights to those observations that were poorly predicted by the previous model (Dietterich, 2000; Zhang and Ma, 2012). In this study, models with top performances were recorded and reported. We also conducted a grid search for the top ten models to fine tune the performance (Zhou, 2012).

We evaluated the model performance based on four matrices, i.e., accuracy classification score, precision, recall, and F1 score. Accuracy is the proportion of correctly predicted objects to the number of total objects. Precision is the actual correct proportion of positive identifications. The recall represents how many of the objects that should have been identified are selected. The F1 score can be interpreted as the weighted average of the precision and recall, namely the harmonic mean of precision and recall. As a result, if either precision or recall is low, the result of F1 will be low.

To estimate the use of BSS versus taxi, we used the best model to predict the likelihood of a person using the BSS system in a given situation. For instance, we created a scenario where a person needs to travel from an origin to a destination census tract for eight kilometers on a weekday with sunny but windy weather. We used the trained model to predict if this person would use BSS or hail a taxi. We create four groups of scenarios to model the influences, with each scenario focused on varying travel distances, times of day, weather conditions, and land use factors on the choice of travel mode, respectively.

4. Results

4.1. Trip patterns

We first explored the average daily trip count and speed for both BSS and taxi from 2014 to 2016 (Table 1). Fig. 1 shows the comparison between the number of BSS and taxi trips. In general, the total trips of the taxi were much higher than those of BSS. BSS trip number was subject to weather conditions and therefore showed a seasonal pattern

Table 1
Temporal variations of trip numbers and speed for BSS and taxi trips. WD and WE stand for weekday and weekend, respectively.

			Night	Morning rush	Morning	Lunch break	Afternoon	Afternoon rush	Evening
Hours			0:00–7:00	7:00–9:00	9:00–11:00	11:00–13:00	13:00–17:00	17:00–19:00	19:00–24:00
BSS	Speed (MPH)	WD	9.54	8.89	8.30	7.81	8.04	8.39	8.38
		WE	8.45	8.66	7.91	7.45	7.23	7.39	7.63
	Count (K)	WD	321.66	1014.02	419.52	466.46	1331.20	1255.50	852.17
		WE	83.27	76.12	197.76	302.19	651.41	272.22	288.70
Taxi	Speed (MPH)	WD	19.80	12.51	12.33	13.89	12.55	10.06	14.34
		WE	15.16	24.37	18.58	15.53	13.55	12.80	13.79
	Count (K)	WD	1163.40	1877.27	2731.09	2648.80	5579.39	3489.68	7079.71
		WE	2669.64	165.14	436.17	724.06	1664.07	1045.47	2650.37

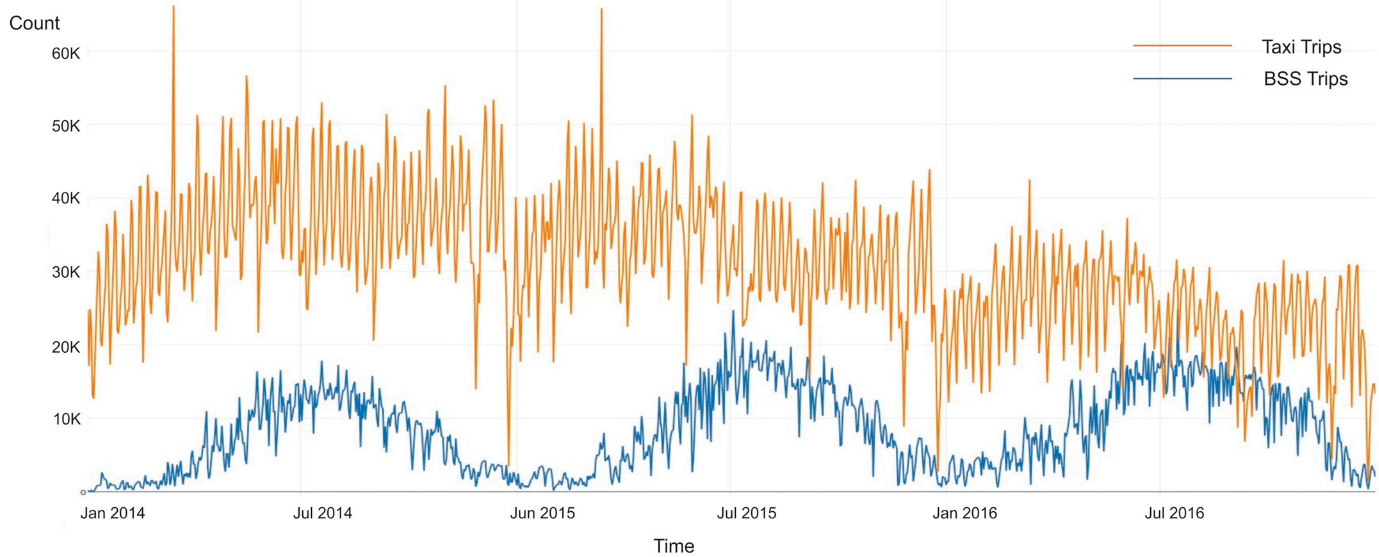

Fig. 1. Daily trip number comparison between BSS and taxi.

with peaks in summers and troughs in winters. The trip number for taxi showed a weaker seasonal pattern but an overall declining trend from 2014 to 2016. Such a declining trend may be largely caused by ride services such as Uber and Lyft. Both BSS and Taxi services revealed fluctuations between weekdays and weekends.

In addition, unlike trip numbers, the trip speed of BSS and taxi showed overall steady patterns. No apparent seasonal trend was observed. The speed of the taxi displayed a much higher variation than that of BSS. On average, taxi speed was much faster than BSS. However, because of considerable variation, there was an apparent overlap between BSS speed and taxi speed (the area where light green and light purple colors overlap) (Fig. 2).

We also explored the trip direction for both BSS and taxi. Fig. 3 shows the dominant directions of BSS and taxi in the morning and afternoon rush hours. The pointing direction of the blue arrows represents the flow direction while the size of the arrow represents the number of trips in this direction. The contour lines represent the logged trip numbers, with red suggesting more trips and green fewer trips. As we can see from the dominant trip directions, there are strong flows into downtown in the morning rush hours for both BSS and taxi trips. During the rush hours in the afternoon, BSS showed a radiating pattern away from downtown. Both BSS and taxi patterns showed high-density trips in the downtown area, but the taxi trips extend out more into the other districts. Based on the contour lines, we found that morning trips showed strong inbound flows, and such flow was more temporally concentrated than the outbound flows in the afternoon.

4.2. Models for travel mode choice

To model the probability of a person using BSS versus taxi under a given spatiotemporal situation, we compared the performance of multiple machine learning models. Due to a large amount of data, we sampled 30,000 objects, with equal numbers of BSS and taxi trips to compare model performance. Table 2 shows the performance of the classification results of different machine learning models. Among the different model families, ensemble models show the best performance,

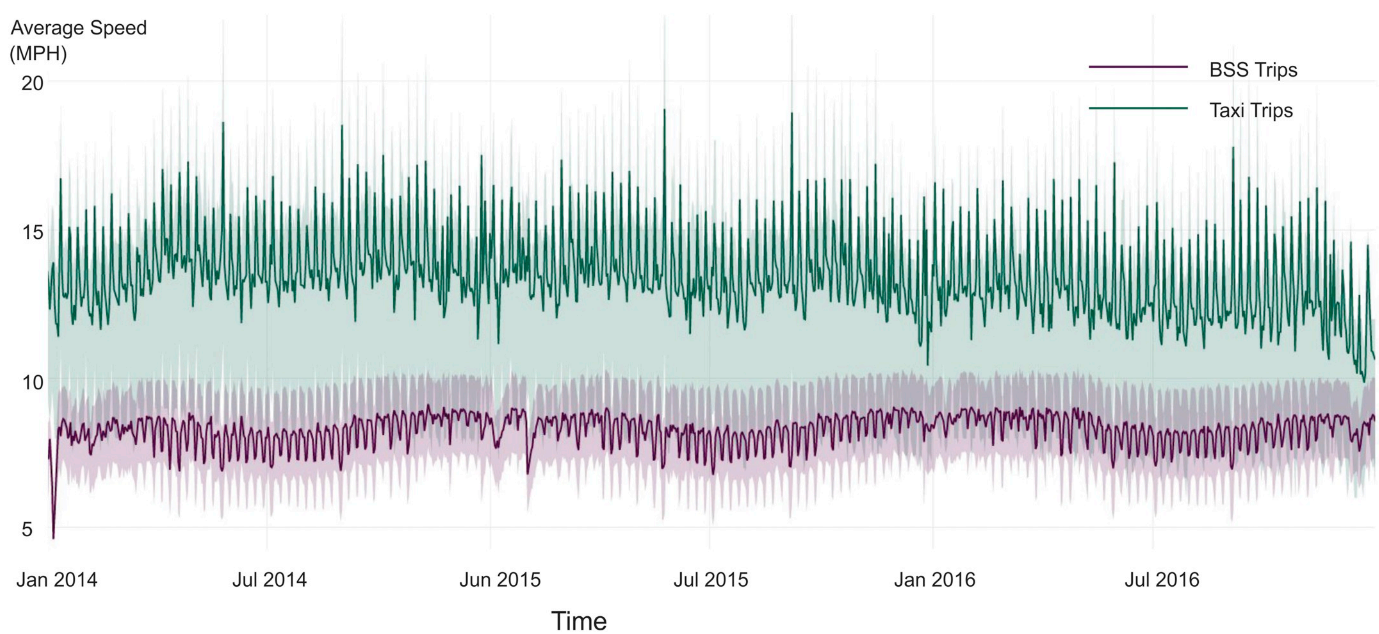

Fig. 2. Daily trip speed comparison between BSS and taxi.

Fig. 3. Dominant trip directions in the morning and afternoon rush hours for BSS and taxi trip.

Table 2
Model performance comparison based on three-fold cross-validation (mean value of three folds cross-validation).

	Logistic	sgd	knn-5	knn-10	knn-20	cart	svc	bayes	ada	bag	rf	et	gbm
Accuracy	0.674	0.671	0.782	0.786	0.792	0.743	0.682	0.747	0.808	0.815	0.820	0.822	0.813
F1	0.666	0.660	0.774	0.766	0.773	0.744	0.697	0.723	0.804	0.811	0.816	0.816	0.808
Precision	0.682	0.683	0.803	0.844	0.848	0.742	0.665	0.797	0.821	0.831	0.837	0.846	0.833
Recall	0.652	0.639	0.746	0.700	0.710	0.745	0.733	0.663	0.788	0.791	0.796	0.788	0.783

Logistic: logistical regression; sgd: regularized linear models with stochastic gradient descent; knn: k-nearest neighbor; cart: Classification and Regression Trees; svc: support vector classification; bayes: Gaussian Naive Bayes; ada: AdaBoost; bag: Bagged Decision Trees; rf: Random Forest; et: Extra Trees; gbm: Gradient Boosting Machines.

Table 3
Comparison between models with the polynomial interaction terms based on three-fold cross-validation (mean value of three folds cross-validation).

	Logistic	sgd	knn-5	knn-10	knn-20	cart	svc	bayes	ada	bag	rf	et	gbm
Accuracy	0.801	0.797	0.784	0.786	0.790	0.736	0.802	0.722	0.807	0.814	0.817	0.819	0.812
F1	0.794	0.792	0.775	0.767	0.771	0.736	0.793	0.675	0.804	0.809	0.812	0.814	0.809
Precision	0.824	0.815	0.807	0.845	0.848	0.735	0.831	0.812	0.816	0.830	0.833	0.838	0.825
Recall	0.767	0.774	0.746	0.702	0.707	0.738	0.758	0.577	0.792	0.790	0.793	0.791	0.793

Logistic: logistical regression; sgd: regularized linear models with stochastic gradient descent; knn: k-nearest neighbor; cart: Classification and Regression Trees; svc: support vector classification; bayes: Gaussian Naive Bayes; ada: AdaBoost; bag: Bagged Decision Trees; rf: Random Forest; et: Extra Trees; gbm: Gradient Boosting Machines.

among ensemble models, random forest, and extra trees achieved the highest scores in all four metrics. Followed by ensemble models, KNN models also generated relatively good results. Linear models, on the other hand, were not able to explain as much variance compared to the rest of the models.

In order to test the interaction between variables, we generated a new feature matrix consisting of all degree-2 polynomial combinations of the features. Table 3 demonstrates the accuracy assessment. After introducing the interaction, the accuracies of linear models such as logistical regression significantly increased. However, the performance of ensemble models generally stayed the same and consistently outperformed the linear models. As a result, we chose not to use interaction models in our final models. We also conducted a grid search for the neural network. We tested parameters including the number of neurons, learning rate, and the number of hidden layers. We searched learning rates including 0.005, 0.01, 0.05, and 0.1. Neuron numbers we tested included 100, 1000, 2000, 5000. Table 4 shows the F1 scores based on training and testing dataset. Overall, when the neuron number increased, the performance of the training dataset improved. However, the accuracy of prediction on the testing dataset improved initially but quickly started to drop. Among the different one hidden layer settings, the learning rate of 0.01 generates the best performance. When we increase the hidden layers to two, over-fitting became more obvious, as the gaps between F1 scores got wider. Finally, the best two-layer model produced an F1 score of 0.796, which was still not as good as the one-layer model.

By comparing different models, we found that the random forest model produced the best results with relatively high processing speed. As a result, we used the random forest to investigate the travel mode choice behavior further. We plotted the number of trees in the forest versus the sample size for the model performance (Fig. 4) and found that once the sample size exceeded 600,000, the performance became stable. In addition, 300 trees in the model generated the lowest classification error. However, the differences in errors between 100 trees and 300 trees were very small. Considering the training time, we sampled 800,000 observations and used 200 trees as the final model to predict the likelihood of using BSS versus taxi.

4.3. Factors important in predicting travel mode choices

We investigated the influences of travel distance, time of the day, weather, and land use factors on the choice of travel mode. By default, we assumed there is a person who needs to travel about 8 km on a weekday in June at noon with a good weather condition. Fig. 6 shows the results of four scenarios with specific varying variables. For instance, Fig. 5a shows the change in the likelihood of using BSS when travel distance increases. People tend to choose BSS when the travel distance is short. Color of the lines represent different months, and intuitively, in winter and spring, people are less likely to use the BSS than in summer and fall. Interestingly, we can observe that the likelihood of choosing BSS drop quickly around six to eight kilometers of travel distance. Such a distance threshold largely corresponds to 25 to 30 min of riding distance. This result may reflect the Divvy Bike policy where the first 30 min of each ride are included in the membership or pass price, but additional fees will apply if each ride exceeds 30 min.

Basically, no matter what season it is, when the travel distance is greater than 8 km, the BSS system becomes much less favorable. Fig. 5b shows the BSS preference in different time of the day and days of the week. In general, BSS is used more during weekdays than the weekend. People tend to use BSS in the daytime rather than during the nighttime. After 6 pm, people tend to select a taxi rather than BSS to travel 8 km. Fig. 5c shows the influence of weather on BSS usage. Precipitation has a major impact on BBS usage. There is a clear trend of decreased BSS use as precipitation increases. The likelihood of using BSS is consistently less than 0.5 during rainy days regardless of the amount of precipitation. In terms of wind, when the wind speed is smaller than 10 mph, it does not have a major impact on USS usage. However, when the speed is greater than 10 mph, people opt to use a taxi rather than BSS. Fig. 5d shows how the built environmental factors are associated with the probability of using BSS. We analyzed the mode choices concerning the number of parks and recreational facilities and the total length (in km) of bike lanes in each census tract. Bike lane length does not have a direct impact on BSS choice, but the number of parks, especially when the park and recreational facility count increases from 15 to 25, people are more likely to use BSS.

Fig. 6 shows the spatial prediction of the likelihood of using BSS at different times of the day. We used the model to predict the probability of people using BSS rather than calling a taxi at 8 am to the south direction and 6 pm to the north direction. In these prediction scenarios, we assumed that this person needs to travel about 8 km on a weekday in June. The weather is good, with a slight wind and no rain. At 8 am, this person is a bit more likely to ride from North Chicago to the downtown

Table 4
F1 scores for different neural network structure.

Learning rate	0.005				0.01				0.05				0.1			
One hidden layer																
Neuron number	100	200	1000	2000	100	200	1000	2000	100	200	1000	2000	100	200	1000	2000
Training	0.808	0.819	0.835	0.834	0.811	0.816	0.819	0.819	0.790	0.792	0.795	0.795	0.785	0.789	0.794	0.789
Testing	0.793	0.795	0.788	0.789	0.798	0.796	0.800	0.797	0.789	0.791	0.794	0.794	0.778	0.786	0.790	0.785
Two hidden layer																
Neuron number	100, 100	200, 200	500, 500		100, 100	200, 200	500, 500		100, 100	200, 200	500, 500					
Training	0.881	0.965	0.982		0.876	0.888	0.888		0.798	0.802	0.804					
Testing	0.765	0.753	0.755		0.774	0.761	0.771		0.793	0.796	0.794					

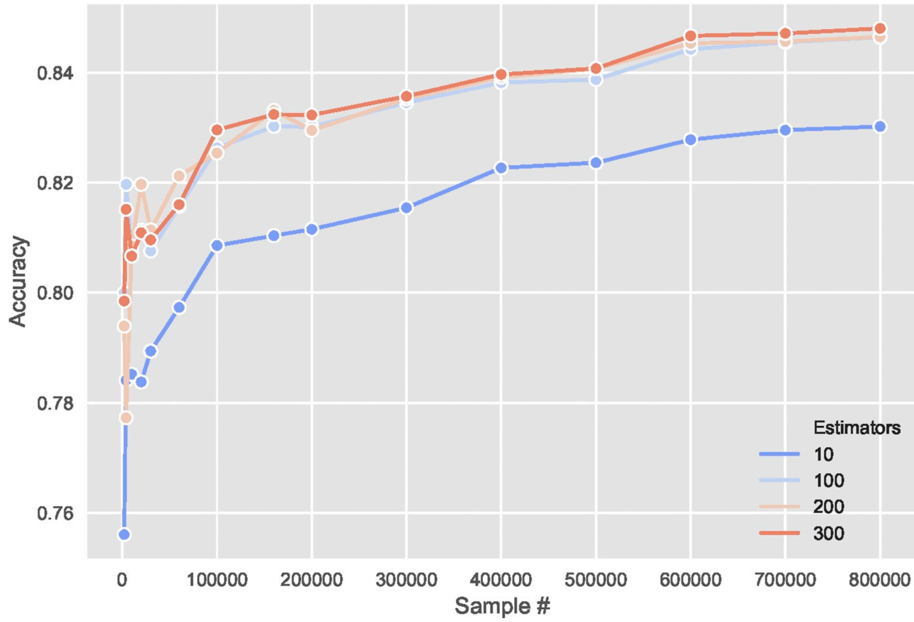

Fig. 4. Accuracy score comparison between different model settings.

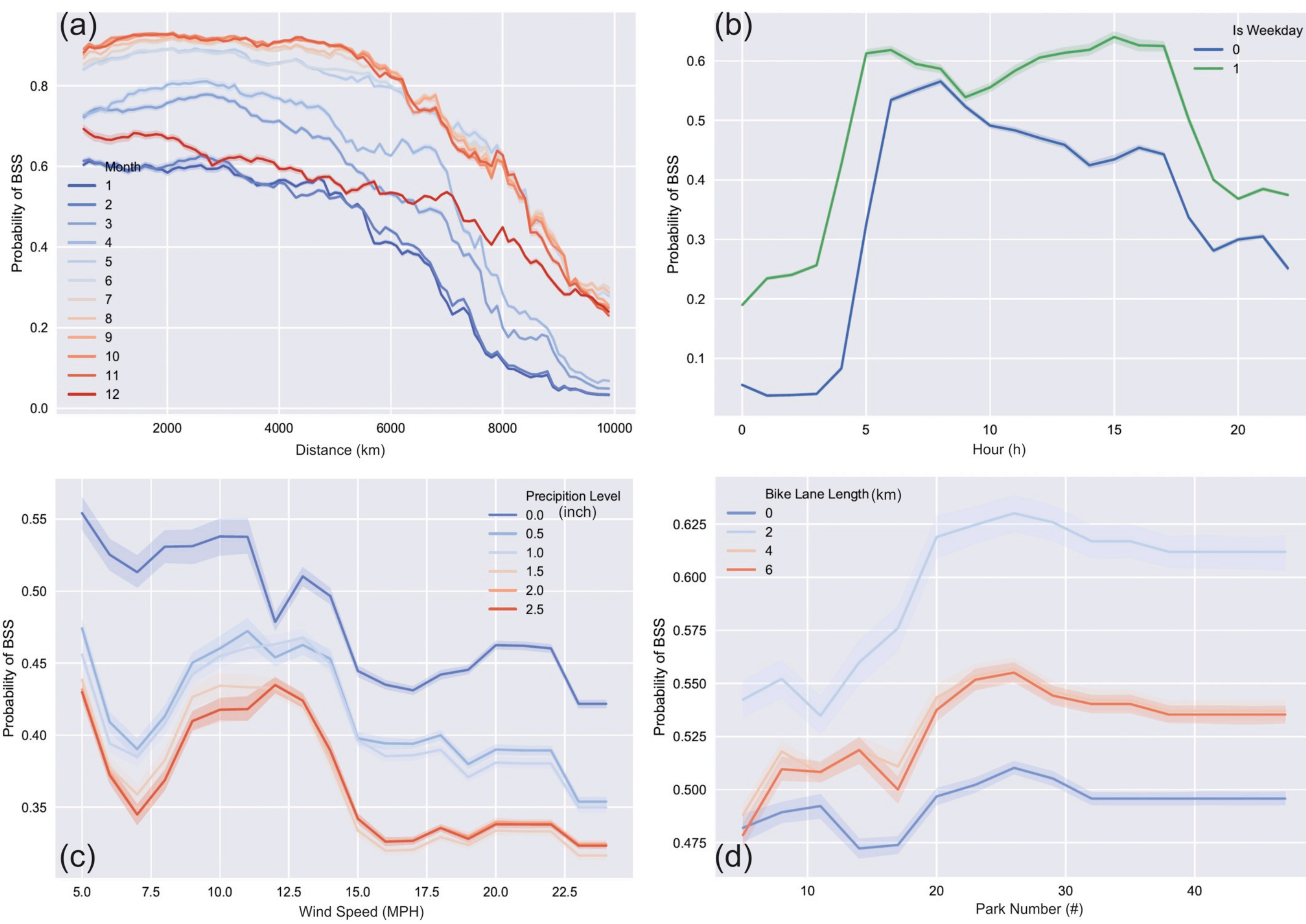

Fig. 5. Probability of using BSS in different conditions. (a) Travel distance and month influences (b) Hour of a day and weekday/weekend influences. (c) Precipitation and wind strength influences (d) Bike lane and park count influence.

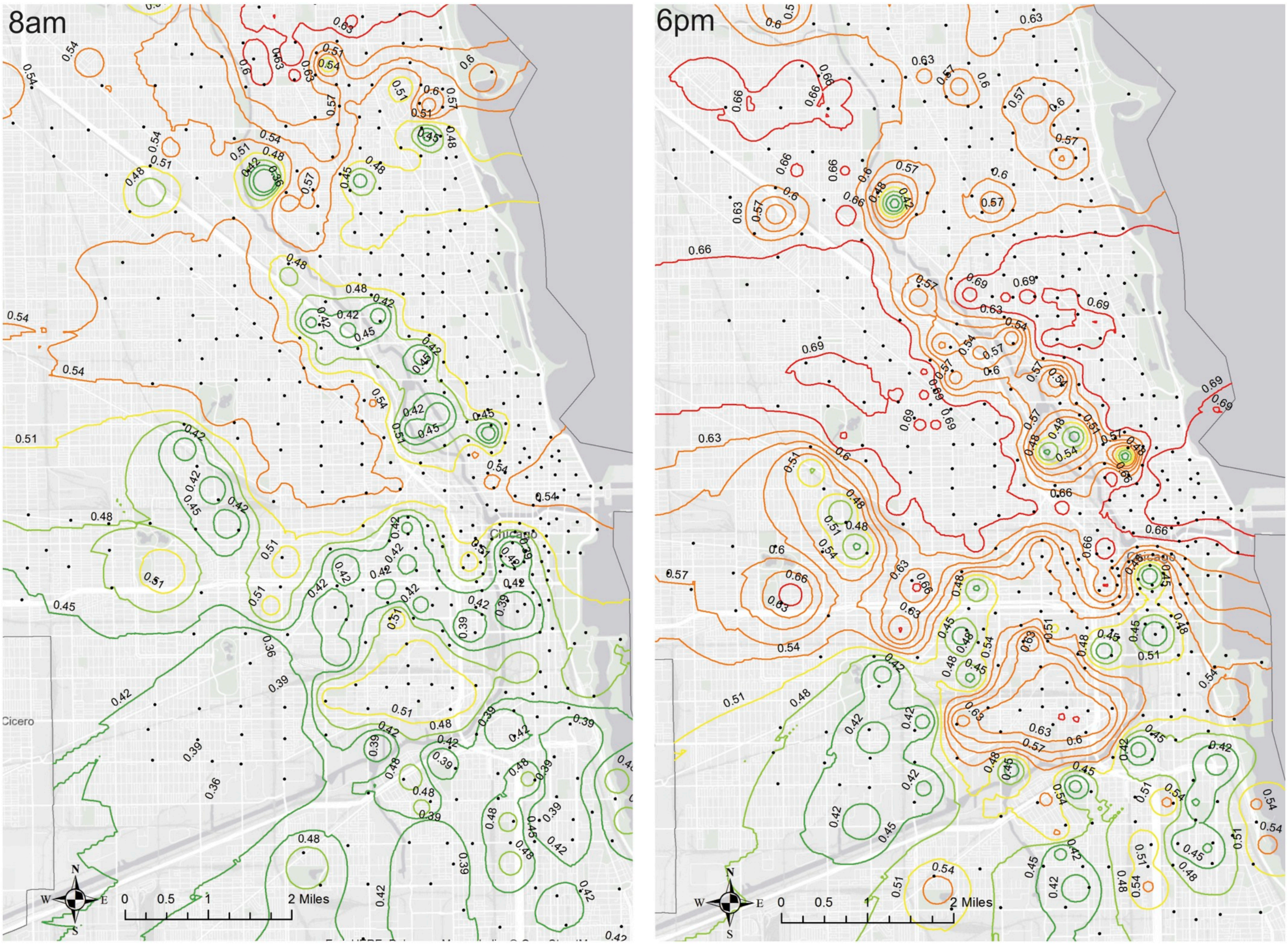

Fig. 6. The spatial prediction of the likelihood of using BSS at a different time of a day.

area. However, in the South of Chicago, this person is more likely to use a taxi. At 6 pm, on the other hand, the likelihood to use BSS becomes higher, especially in the CBD areas. The demand for BSS becomes higher in the south of Chicago compared to the demand in the morning.

5. Discussions and concluding remarks

Taxis provide convenient door-to-door services at any time of day in big cities. In the recent decade, the growing number of BSS provides critical alternative means of transportation in many cities. This study explored the spatiotemporal distributions of trip patterns of these two essential means of transportation in the City of Chicago. We applied machine learning techniques to predict the optimal travel mode choice between BSS and taxi.

We modeled travel mode choices between BSS and taxi at the individual origin-destination pair level. We found key factors predicting people's decision to use BSS over the taxi. First, we found that environmental factors, such as the density of parks, were essential indicators of using BSS, suggesting that bikers appreciate recreational opportunities and aesthetic qualities along the way. Second, consistent with previous findings related to bike sharing usage (Gebhart and Noland, 2014; Mattson and Godavarthy, 2017), we found that seasonal and weather conditions, including air temperature, precipitation, and wind speed, were important predictors for mode choices. Specifically, during cold winter days, or when it was rainy or windy, people were significantly more likely to use a taxi. Third, the results demonstrated that the distance between origin and destination and the trip direction were essential factors in explaining travel mode choices. The 6–8-km distance seemed to be a consistent threshold, beyond which people's preferences for taxi increased primarily. One previous study of BSS trips alone conducted in Cork City, Ireland, identified that three-quarters of trips were made at a distance below 1850 m (Caulfield et al., 2017). The different findings regarding the distance threshold could be attributable to the fact that our study aimed to identify the relative strengths of BSS versus taxi. In this sense, BSS remains competitive up to 6–8 km. Another explanation would be related to the different sizes and densities of the two cities, as well as the different BSS program policies. We also observed salient temporal patterns in travel mode choices. For example, during morning rush hour, people who enter the CBD from North Chicago were more likely to use BSS than those traveling to CBD from South Chicago, which could be an indicator of differences in infrastructure, environmental quality, or perceived safety. Furthermore, our finding of travel patterns resonated the current Divvy Bike pricing strategy of a flat rate in the first 30 min (of both subscribed membership and single pass). These findings advance our knowledge of the spatial and temporal determinants of travel mode choices and offer critical transportation planning implications. According to the National Association of City Transportation Officials (NACTO), an association of 68 major North American cities and 11 transit agencies, there are more than 84 million trips taken via public shared bikes and scooter in 2018, which is more than twice the number of those in the previous year (https://nacto.org/shared-micromobility-2018/). We would expect the number to increase continuously. As a result, understanding travel patterns and choices will become increasingly relevant for cities in the coming years.

When modeling the travel choice between BSS and taxi, we have considered a suite of machine learning techniques. While generally non-linear algorithms performed better than the linear ones, we found the prediction accuracy of a simple logistical regression increased mostly when including the polynomial terms of features. Moreover, the performance of multi-layer neural network did not surpass that of classical non-linear models (e.g., random forest, Tables 2, 3, and 4). Within the neuron network family, a deep neuron network only increased the prediction accuracy at a marginal rate (Table 4). Therefore, the model selection process also demonstrated the potential tradeoff between performance and computational cost in travel choice modeling.

There are several limitations to this study, which also pave the paths for future research. First, as with many other previous studies that model BSS demands, modeling at a fine spatial level is challenging (Faghih-Imani and Eluru, 2016). In this study, prediction accuracies were limited because trips of BSS and taxi were aggregated to the census tract levels. Variations exist concerning trip origins, destinations, and the resulting environmental conditions within each census tract. Hence, finer-scaled datasets that include the actual geolocations of the origin and destinations would reveal more accurate environment-travel mode choice relationships. Second, some census tracts show a smaller number of taxi trips. For privacy reasons, if a trip's origin and destination census tracts showed less than three total trips in the 15-min window, it was removed from the dataset. Such a procedure reduced the number of observation and introduced a bias related to specific demographic and environmental conditions. Third, because the BSS dataset included the origins and destinations but not the exact route between the origins and the destinations, the distance and speed were estimates rather than the actual speed of travel. We used Google Direction API to find the closest biking routes between origin and destinations in order to estimate the travel speed, which was the best estimation we could achieve. In future studies, data collected with GPS tracks of the trips would enhance the accuracy of the dataset and allow better model predictions. Fourth, this study considers the choice of travel between BSS and taxi. In the future, more modes of transportation can be incorporated into the analytical framework, e.g., public transit and ride services (Zhou et al., 2017; Wang and Mu, 2018). Finally, this study used data from one city during a three-year window. However, trip distribution and preference in urban areas have been demonstrated to be highly dependent on the urban form. This study uses data from Chicago as a testbed for the modeling approaches. Although the results show great potential in using machine learning to predict transport choices, the findings need to be applied in other geographic areas and urban contexts with caution. Future study should examine a broader range of urban settings and test the sensitivity of the findings to variations in urban configurations.

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